

EE-482: Network Security and Cryptography

LECTURE NO 11: *Symmetric and Asymmetric Cryptography*

Engr. Tehseen Ahsan

**Assistant Professor, Computer Engineering Department
HITEC University Taxila Cantt, Pakistan**

Organization of Lecture

1. Introduction
2. Symmetric Versus Asymmetric Cryptography
3. The best of both worlds

1. Introduction

↳ Asymmetric cryptography i-e using receiver's public key for encryption solves the problem of key agreement and key exchange

2. Symmetric Versus Asymmetric Cryptography

<u>Characteristics</u>	<u>Symmetric</u>	<u>Asymmetric</u>
1) Key used for encryption/decryption	Same key used	One key for E & other key for D.
2) Speed of E/D	Very fast	Slower
3) Size of encrypted Text	Usually same or less than original plain text	More than original plain text
4) Key agreement/Key exchange	A major problem	No problem
5) No. of keys required as compared to no. of Participants in message exchange	Equal to about the square of no. of participants (a large no.)	Same as no. of participants
6) Usage	Encryption/Decryption, Cannot be used for digital signatures	E/D as well as digital signatures

3. The Best of Both Worlds (1/7)

* The best of both worlds

objectives:

- 1) Solution should be completely secure
- 2) Encryption & Decryption process shouldn't take a long time.
- 3) Generated cipher text should be compact in size.
- 4) Solution should extend to a large no. of users without any additional complications.
- 5) Key distribution problem should be solved.

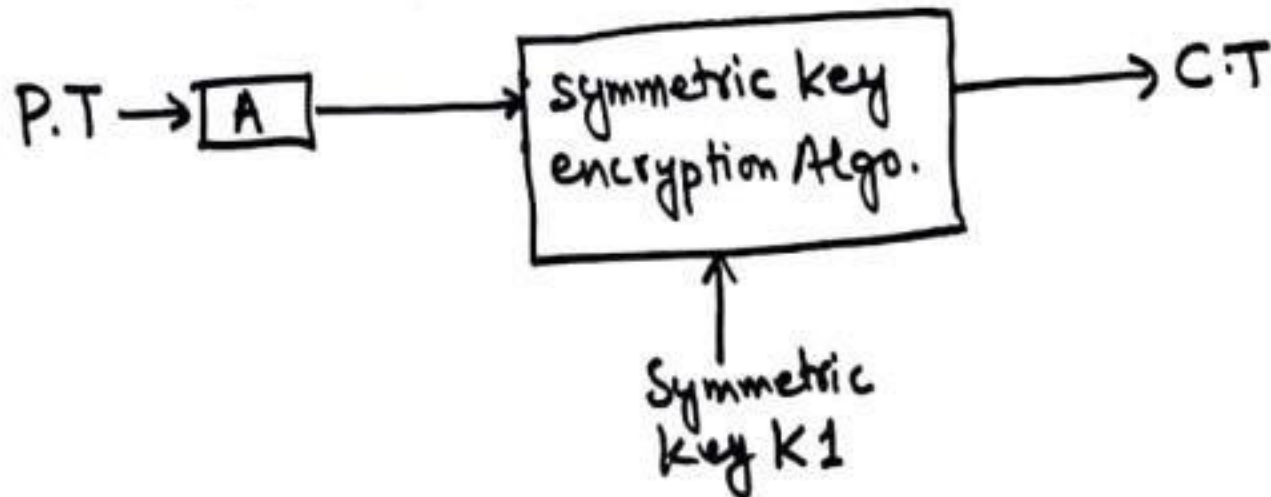
Solution: Combine the best of symmetric & asymmetric key cryptography

3. The Best of Both Worlds (2/7)

step 1: Sender A encrypts original P.T using symmetric key algorithm (DES, IDEA etc).
output is cipher text C.T

Key used for this operation K_1 called one-time symmetric key

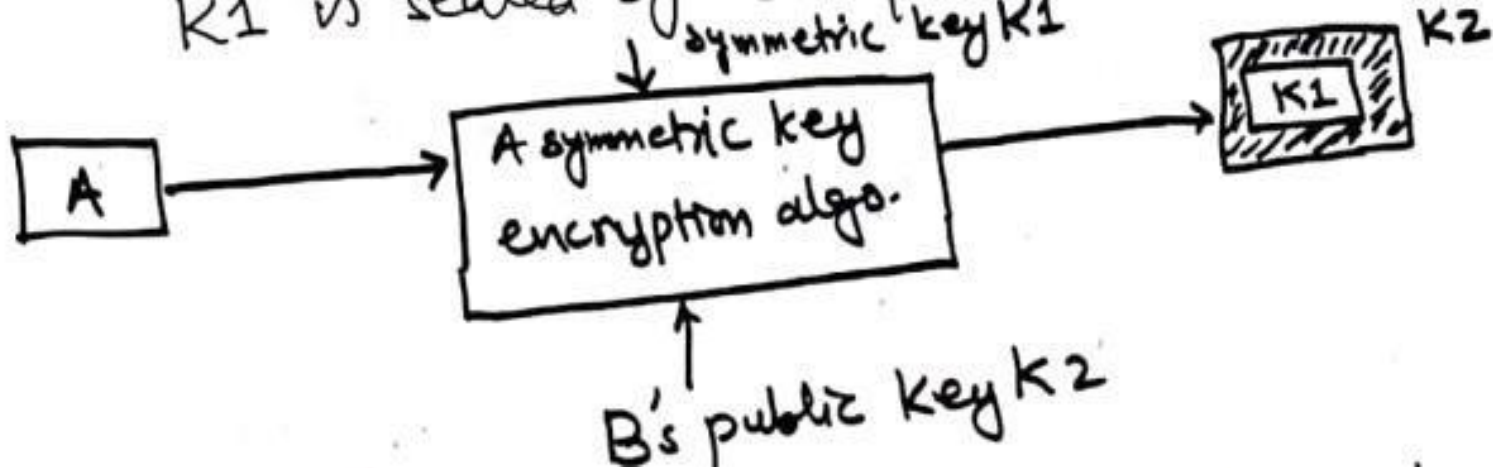
K_1 is used once & then discarded



3. The Best of Both Worlds (3/7)

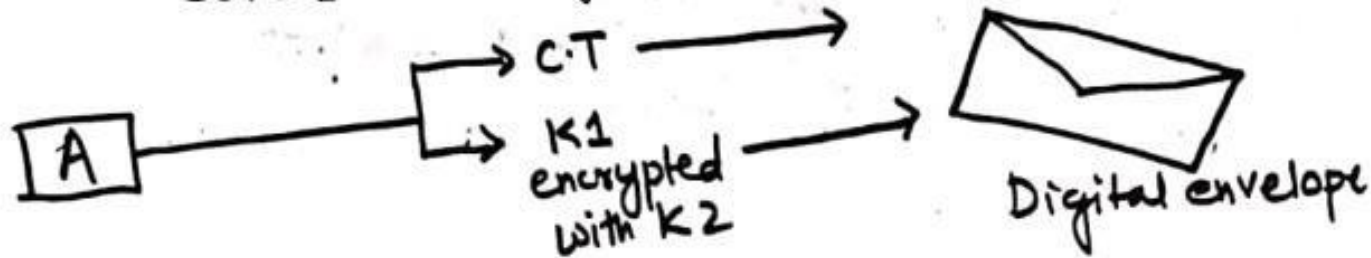
Step 2: A takes the one-time symmetric key K_1 and encrypts it with B's public key K_2 . This process is called key wrapping of symmetric key.

K_1 is sealed by B's public key K_2 .



3. The Best of Both Worlds (4/7)

Step 3: A puts the C.T & encrypted symmetric key K1 together inside a digital envelope & sends the digital envelope over the network

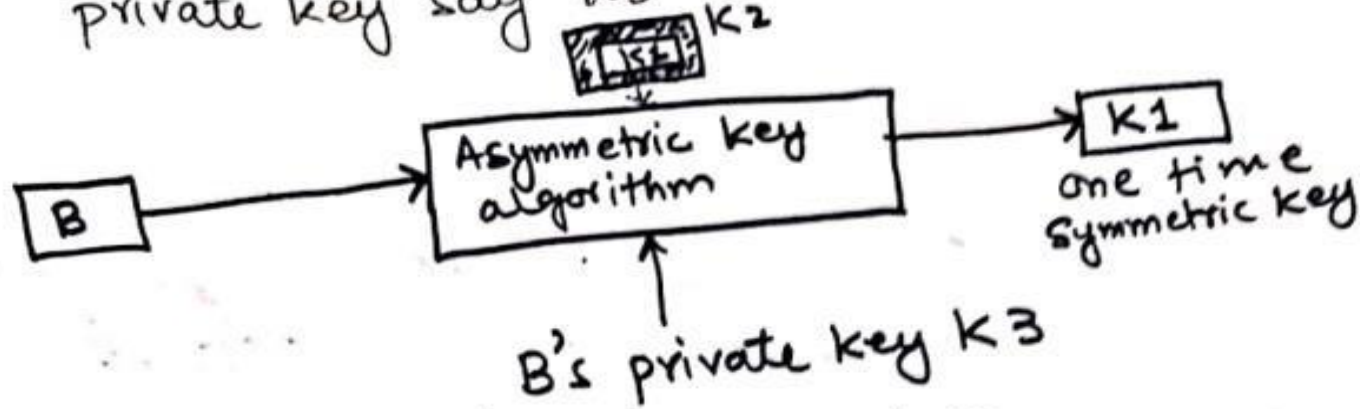


Step 4: B receives the digital envelope & receives C.T and one time session key encrypted with B's public key (K2). $\rightarrow K1$

Step 5: B uses same asymmetric key algorithm as used by A (e.g. RSA) in step 2 to decrypt the logical box containing K1 encrypted with K2.

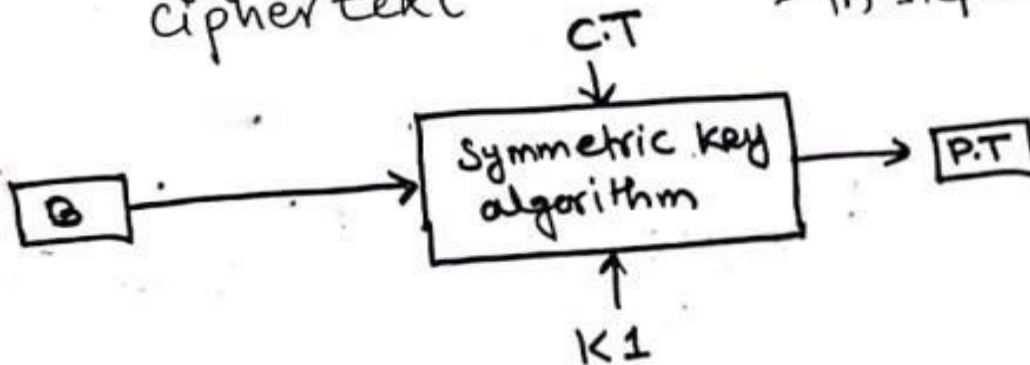
3. The Best of Both Worlds (5/7)

B will only be able to decrypt using his private key say K_3



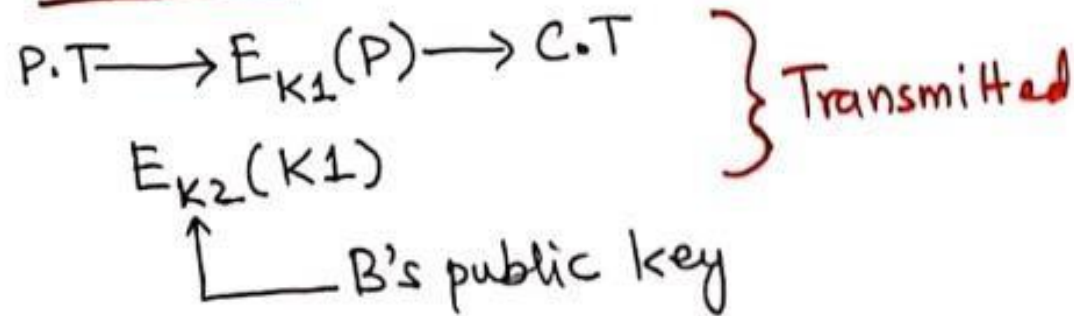
B public/private pair = K_2/K_3

Step 6: Finally B applies same symmetric key algo. as used by A & K_1 to decrypt ciphertext → in step 1

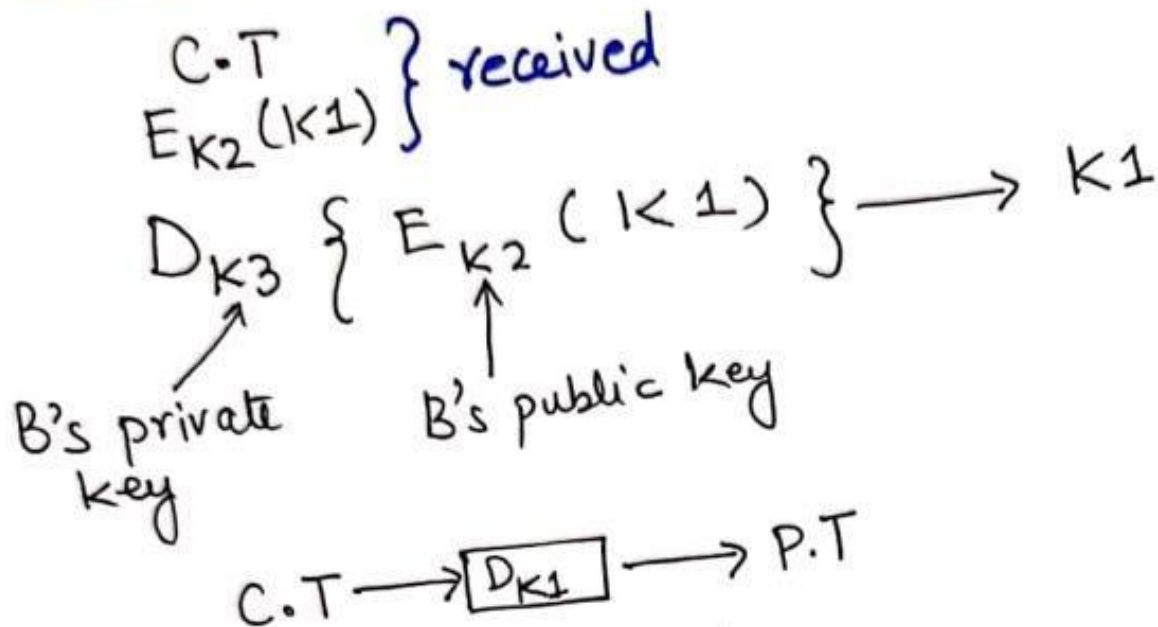


3. The Best of Both Worlds (6/7)

Sender (A)



Receiver (B)



3. The Best of Both Worlds (7/7)

- Encryption with K_1 is fast.
- C.T is same as P.T
- Size of K_1 is 56 bits or 64 bits. (small) & encryption using symmetric key is fast.
- Solved the problem of key exchange

References: Textbooks

1. *“Cryptography and Network Security”, by Behrouz A. Forouzan, Debdeep Mukhopadhyay, 2nd Edition, McGraw Hill Education (India) Private Limited.*
2. *“Cryptography and Network Security Principles and Practice”, by William Stallings, 6th Edition, Pearson.*
3. *“Computer Security Principles and Practice”, by William Stallings, Lawrie Brown, 4th Edition, Pearson.*